13 October 2026

Professor Zita Vale
Editor-in-Chief
Applied Energy

Dear Professor Vale,

Please consider our substantially revised manuscript, “Characterisation and Quantification of Data Centre Flexibility for Power System Support”, for publication as a research article in Applied Energy. This is a resubmission following rejection of manuscript APEN-D-25-21553. We refer explicitly to that earlier submission in accordance with your decision letter. The author order differs from the previous submission: James Day is now first author, followed by Mehmet Türker Takci and Meysam Qadrdan. All authors have agreed to this order.

The study develops a whole-facility framework for quantifying the magnitude and feasible duration of data centre flexibility through coordinated IT scheduling, UPS batteries and cooling. Its contribution is the assessment of grid-facing responses from a common operating baseline, with operational constraints and comparative post-event recovery conditions. This connects energy management within data centres to the practical characterisation of demand-side flexibility for power system support.

The revision substantially expands the evidence from a single-day cost comparison to linked daily optimisation over 2025, with fourteen annual scenario chains and sensitivities to workload flexibility and storage capacity. The central annual electricity cost saving is 5.38%; this supersedes the 10.02% single-day estimate in the previous submission and is not a like-for-like comparison with it. We provide a component cost breakdown, an explicit optimisation workflow, clearer workload-input provenance, detailed solver and numerical checks, and revised figures. We also distinguish certified feasible durations from unproven maximum durations and clarify the limits of the network and economic interpretations.

We enclose a clean revised manuscript, a copy with changes marked against the rejected version, and point-by-point responses to all comments from Reviewers 2 and 5. The submission also includes highlights, supplementary information and the declaration of interests.

All authors have approved this version and its submission. The work is original, has not been published previously and is not under consideration elsewhere. The authors declare no competing interests. Funding, author contributions and the use of ChatGPT Codex and Claude Code for manuscript preparation and coding are disclosed in the manuscript. I remain the corresponding author.

Thank you for considering this revised submission.

Yours sincerely,

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